

EMBEDDED SYSTEM AND DESIGN

EXPERIMENT NO. 1

SYMPHONY OF LIGHTS

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DISCUSSION

The symphony of lights is an enthralling event with its appealing combination of technology, creativity, and urban landscape. Against the atmosphere of a lively cityscape, the Symphony of lights transforms buildings into canvas-like screens, painting stories of culture and invention through synchronized lighting. With that said, as a student of Embedded System and Design, the Symphony of lights was recreated through the use of Arduino uno, 9 LEDs, and 5 different parts of musical pieces.

The objectives of the experiment were: (1) To be able to demonstrate how an Arduino Kit will be

used to implement the Symphony of lights; (2) to be able to demonstrate that the unique chosen music can be synchronized with the series of LEDs; (3) to be able to apply learnings from Microprocessor and Logic Circuits and Switching Theory; and (4) to be able to apply an embedded system in a simple handshaking using an Arduino Kit. The materials used for this experiment are the Arduino Kit, 9 LEDs, and 4 switch buttons.

The circuit construction was simple since the LEDs were parallel, for the outputs should be independent. Moreover, the connections were straightforward. The Arduino UNO has 14 digital pins (0-13), and 9 pins were used for the experiment (5-13). Generally, the most used function is digital Write, which provides a pin value of HIGH or LOW,

ON or OFF respectively. Another is analogWrite, which can be used in gradual changes in light intensity. The most important cog or element that connects to other features of the experiment is the timing of the song with the LEDs through the use of the delay function. With Arduino Uno, there is no built-in function that assures the timing of the LEDs with a song; hence, the timing was and should be manually estimated. Along with the timing of the LEDs, the rhythm and melody should be considered, and as for the rhythm, one can use the beats for the base, instruments, and many more. So, as those were considered, different patterns emerged. LEDs wave from left to right, right to left, in to out, out to in, odd or even, blinking of lights, and many more. Since different songs have varied rhythms, more patterns of state-changes gradually produced the symphony of lights when combined.

CONCLUSION

The first experiment, Symphony of Lights, primarily aimed to produce a system that synchronizes light and sound. In the process, an Arduino Kit was utilized for its connections, arduino uno, 9 LEDs, and 4 switch buttons to implement the experiment. 5 songs, namely Raining in Manila by Lola Amour, Zenzenzense by Radwimps, Usseewa by Ado, Sparkle by Radwimps, and the Littleroot Town Theme Song, were successfully synchronized with the LEDs, and all were according to their timing and beat. With the changing state of the LEDs, application of the circuit, and connecting wires, the lessons in Microprocessor and Logic Circuits and Switching Theory were applied adequately. Lastly, an embedded system, which is the symphony of lights as

a whole, was satisfactorily created. Therefore, all objectives for experiment 1 were met.